Day 54

"CLOUD SECURITY"

Introduction to Cloud Forensics:

Cloud forensics is the application of digital forensic investigation in cloud environments, focusing on identifying, collecting, and analyzing data while preserving its integrity. Unlike traditional forensics, cloud forensics varies with service and deployment models. In SaaS, investigators rely on the CSP for application logs, while in IaaS, they can directly access virtual disks for analysis. In private clouds, physical access to evidence is possible, but in public SaaS models, investigations depend heavily on logs and audit reports provided by the CSP.

Uses of Cloud Forensics

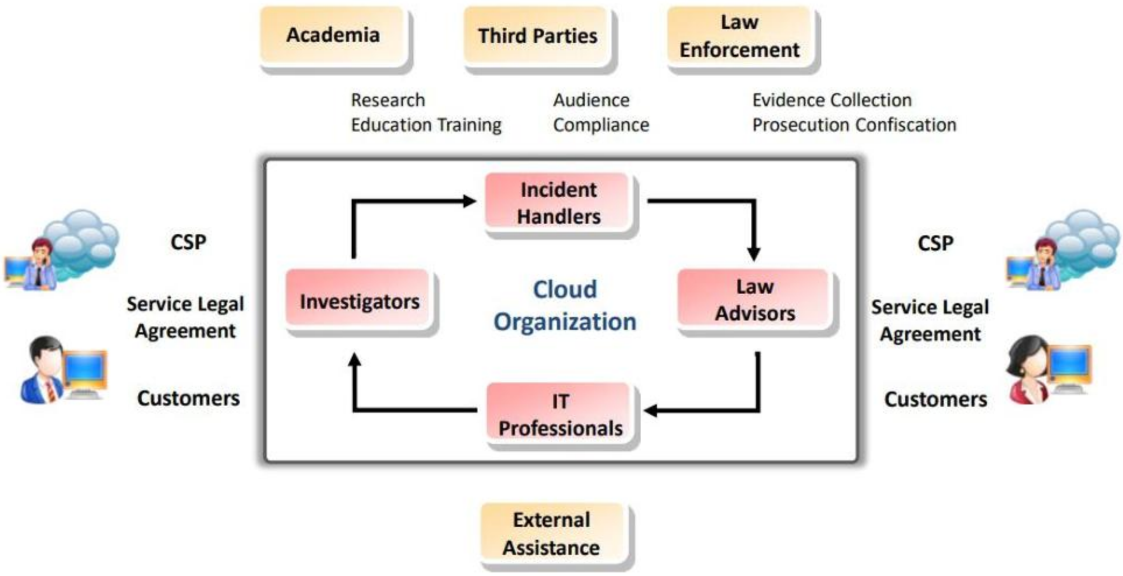
- Investigation: Helps track organized cybercrime, policy violations, and suspicious activities in the cloud.
- Troubleshooting: Resolves functional, operational, and security issues in the cloud environment.
- Log Monitoring: Collects, examines, and correlates logs across endpoints, supporting auditing, compliance, and due diligence.
- Data & System Recovery: Recovers deleted or encrypted data and restores affected systems after attacks.
- Compliance: Ensures organizations follow due diligence, protect critical data, maintain audit records, and notify affected parties of data exposure.

Cloud Crimes

Cloud crimes are criminal activities where the cloud is used as a subject, object, or tool:

- Cloud as a Subject: Crime occurs within the cloud environment, such as identity theft, unauthorized data modification, or malware injection.
- Cloud as an Object: The CSP itself is targeted, e.g., DDoS attacks aimed at disrupting cloud services.
- Cloud as a Tool: The cloud is used to commit crimes against others, such as launching attacks from compromised accounts or storing/sharing crime-related evidence.

Cloud Forensics: Stakeholders and their roles:



Cloud Forensics Challenges- Architecture and Identification

Challenge	Description
Deletion in the cloud	• The total volume of data and users operating regularly in a cloud ecosystem confines the number of backups the CSP will retain • CSPs may not implement the necessary methods to retrieve information on deleted data in an IaaS or PaaS delivery model
Recovering overwritten data	• It is difficult to recover the data marked as deleted because it may get overwritten by another user that shares the same cloud
Interoperability issues among CSPs	• The collection and preservation of forensic evidence is challenging because there is lack of interoperability between CSPs and lack of control from the consumer's end regarding the proprietary architecture and/or the technology used
Single points of failure	• The cloud ecosystem has single points of failure, which may have adverse impacts on the evidence acquisition process
Criminals can access low cost computing power	• Cloud computing provides computing power that would otherwise be not available to criminals at a low budget, thus letting unpredictable attacks that would be unfeasible outside a cloud environment
Real-time investigation via intelligent process is not possible	• Investigating real-time incidents in the cloud is difficult because it requires an intelligence process, which is often not possible while working along with the CSPs or other actors; additionally, a special legal measure must be applied in many cases to collect data
Malicious code may circumvent VM isolation methods	• Vulnerabilities in server virtualization allow malicious codes to evade the VM isolation methods and interfere with other guest VMs or the hypervisor
Multiple venues and geo-locations	• Managing the scope of data collection is challenging because distributed data collection and chain of custody from multiple venues or unknown geo-locations can cause various jurisdictional issues
Lack of transparency	• The operational details of a cloud are not apparent to investigators; this results in lack of trust and difficulties in auditing
Criminals can hide in cloud	• The distributed nature of cloud computing allows criminal organizations to maintain isolated cells of operation to preserve the anonymity of each cell from others, thus it may be difficult for investigators to identify and correlate the cells
Cloud confiscation and resource seizure	• Cloud confiscation and resource seizure may often affect the business continuity of other tenants
Errors in cloud management portal configurations	• Configuration errors in cloud management portals may allow an attacker to gain control, reconfigure, or delete another cloud consumer's resources or applications • It is difficult to find the source of such unauthorized change because the cloud management portal is simultaneously used by multiple tenants
Potential evidence segregation	• The segregation of potential evidence pertaining to a tenant in a multi-tenant cloud system is a challenge because there are no technologies that do this without breaching the confidentiality of other tenants
Boundaries	• Protecting system boundaries is challenging because it is difficult to define system interfaces
Secure provenance	• It is a challenge for investigators to maintain a proper chain of custody and security of data, metadata, and possibly, the hardware because it is difficult to determine the ownership, custody, or exact location
Data chain of custody	• It is probably impossible to identify and validate a data chain of custody owing to the multi-layered and distributed nature of cloud computing

<u>Cloud Forensics Challenges- Data Collection</u>

Challenge	Description
Decreased access and data control	• In each combination of cloud service and deployment model, the investigator encounters the challenge of limited access and control to the forensic data
Chain of dependencies	• The CSPs and most of the cloud applications often rely on other CSP(s), and the dependencies in a chain of CSP(s)/client(s) can be prominently dynamic • In such conditions, cloud investigation may depend on the examination of each link in the chain and the level of complexity of the dependencies
Locating evidence	• Locating and collecting evidence is a challenge because data in cloud may be quickly altered or lost and there is limited knowledge regarding where or how data is stored in the cloud
Data Location	• Collecting data from the target is challenging because it is stored in different data centers and geographic regions
Imaging and isolating data	• Data imaging and isolating a migrating data target is challenging in the cloud ecosystem owing to its key characteristics: elasticity, automatic provisioning/deprovisioning of resources, redundancy, and multi-tenancy
Data available for a limited time	• Data collection and preservation of VM instances are challenging tasks owing to insufficient standard practices and tools
Locating storage media	• It is difficult to locate the storage media in a cloud ecosystem because it requires an in-depth understanding of the cloud architecture and implementation
Evidence identification	• Evidence identification is challenging because the sources/traces of evidence are either not accessible or created or stored differently in comparison to those in non-cloud environments
Dynamic storage	• Often, CSPs dynamically allocate storage based on the consumer's request. This complicates the data collection process during investigation
Live forensics	• Validating the integrity of the collected data is challenging because the data within a cloud system are volatile and frequently changing. Additionally, live forensics tools may make modifications to the suspected system
Resource abstraction	• Identifying and collecting evidentiary data is challenging because the resources are abstracted and the information about cloud architecture, hardware, hypervisor, and file system type is not available to understand the cloud environment
Application details are not available	• Obtaining details of cloud-based software/applications used to create records is challenging because such details are usually unavailable to the investigator
Additional collection is often infeasible in the cloud	• Collecting additional evidence is often unfeasible in a cloud system because specific data locations are not known, the sizes may be huge, and non-standard protocols and mechanisms may be used to exchange data; additionally, they may be poorly documented or not documented
Imaging the cloud	• Imaging the cloud is a challenge because it is unfeasible; moreover, while partial imaging may have legal consequences in the presentation to the court
Selective data acquisition	• Selective data acquisition in the cloud is a challenge because it requires prior knowledge about the relevant data sources, which is very difficult
Cryptographic key management	• Data decryption is a challenging task because ineffective cryptographic key management can result in losing the ability to decrypt the forensic data stored in the cloud
Ambiguous trust boundaries	• In a multi-tenant cloud environment, using cloud services may increase the data integrity risks at rest and during processing • Not all CSPs implement vertical isolation for the client data that leads to questionable data integrity
Data integrity and evidence preservation	• For stakeholders, maintaining evidence quality, evidence admissibility, data integrity, and evidence preservation is challenging because the faults and failures in data integrity are shared among multiple actors, and the chance for such faults and failures is higher in the cloud environment due to sharing of data/responsibilities
Root of trust	• Determining the reliability and integrity of cloud forensics data is a challenge due to the dependence on the collective integrity of multiple layers of abstraction in the cloud system

Cloud Forensics Challenges- Logs

Challenge	Description
Decentralization of Logs	Log information is not stored on any single centralized log server in the cloud; it is decentralized among many servers.
Evaporation of Logs	Some logs in the cloud environment are volatile; for example, VMs. When a VM instance is shut down, its logs vanish.
Multiple Layers and Tiers	There are many layers and tiers in the cloud architecture and logs are generated in each tier, which are valuable to the investigator; however, collecting them from different locations is challenging (e.g., application, network, operating system, and database)
Less Evidentiary Value of Logs	Different CSPs and different layers of cloud architecture provide logs in different formats (heterogeneous formats) and not all logs provide crucial information for forensic investigation purpose; for example, who, when, where, and why some incident was executed

Cloud Forensics Challenges- Legal

Challenge	Description
Missing terms in contract or SLA	• Lack of forensic related terms in cloud contracts is challenging because it can prevent the generation and collection of the existing appropriate data as well as the generation of potentially appropriate data
Limited investigative power	• In civil cases, investigators are often provided with limited investigative power to properly obtain data under the respective jurisdictions
Reliance on cloud providers	• Acquiring forensic data from the cloud is challenging because it requires the cooperation of CSPs, which may be limited by the number of employees and other resources at the provider end
Physical data location	• Specifying the physical location(s) of data on a subpoena is challenging because the requestor often does not know where the data is physically stored
Port protection	• Scanning ports is challenging because CSPs do not provide access to the physical infrastructure of their networks
Transfer protocol	• Dumping the TCP/IP network traffic is a challenge because the CSPs do not provide access to the physical infrastructure of their networks
E-Discovery	• Response time for e-discovery is challenging owing to the ambiguity of data location and uncertainty about whether all relevant data were discovered
Lack of international agreements and laws	• Gaining access to and exchanging data are challenging owing to the lack of international collaboration and different legislative processes in different nations
International cloud services	• Real-time, live access to data on international cloud services is challenging owing to the lack of definition on the scope of data acquisition on non-national cloud services and agreements dealing with the authority to access the data
Jurisdiction	• Gaining legal access to the data is challenging because international jurisdiction guidelines have not been identified
International communication	• Achieving effective, timely, and efficient international communication when dealing with an investigation in a multi-jurisdictional cloud is challenging because the existing mechanisms and networks for such communication are often slow and inefficient
Confidentiality and Personally Identifiable Information (PII)	• Preserving the privacy of personal, business, and governmental information in cloud is challenging owing to the lack of legislation governing the conditions under which such data can be accessed by investigators
Reputation fate sharing	• For CSPs and co-tenants, recovering the reputation affected by the illegal activities of some cloud consumers is challenging because spammers may get use the IP range of the CSP to blacklist these IP addresses • This could potentially disrupt the service of legitimate cloud customers if they are assigned with blacklisted IP addresses

Cloud Forensics Challenges- Analysis

Challenge	Description
Evidence correlation	• Correlation of an activity across multiple CSPs is a challenge owing to the lack of interoperability
Reconstructing virtual storage	• Virtual storage media duplication in some cloud ecosystems may cause damage to the actual media, thereby adding the risk of being prosecuted • Additionally, reconstructed algorithms must be developed and verified
Timestamp synchronization	• Correlating the activities observed with accurate time synchronization is a challenge because the timestamps may be inconsistent between different sources
Log format unification	• The unification/conversion of different log formats to a consistent one is challenging owing to the enormous resources available in the cloud. This may also result in the lack and/or exclusion of critical data • In contrast, uncommon or proprietary log formats of one party can become a major obstacle during investigations
Use of metadata	• Using metadata as an authentication method may lead to risks because the common fields (creation date, last accessed date, last modified date) may change when the data are transferred to and from the cloud or during the data collection process • Consider the impact of cloud on metadata and check if the CSP preserves metadata and is readily accessible for e-discovery purposes
Log capture	• Timeline analysis of logs for the DHCP log data is a challenge because the log data collection method of different CSPs is different

--The End--